

Laboratory Procedure 101

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Basic Laboratory practices

1 Good Laboratory protocols

- Be punctual and attentive during laboratory sessions.
- **Handle all instruments and apparatus with care.**
- Record observations neatly, with correct units and significant figures.
- Use scientific notation where appropriate for very large or very small numbers.
- **Always mention the scale** in the legend while plotting graphs.
- **Ensure safety protocols are followed at all times.**

2 Scientific Notation

In laboratory work, measured quantities often span a wide range of magnitudes. To express such values clearly and concisely, we use **scientific notation**. In this system, a number is written in the form:

$$N = a \times 10^n$$

where a is a real number such that $1 \leq |a| < 10$, and n is an integer representing the order of magnitude.

Advantages of Scientific Notation

- It provides a compact and standardized way of expressing very large or very small numbers.
- It makes comparison of orders of magnitude straightforward.
- It reduces errors in recording and interpreting experimental data.
- It simplifies calculations involving multiplication and division.

Rules for Usage

1. Always express the coefficient a with the appropriate number of **significant figures** reflecting the precision of the measurement. For example:

$$0.000\,345 \rightarrow 3.45 \times 10^{-4}$$

2. Use scientific notation consistently in data tables, graphs, and calculations.
3. Units must always be retained when expressing values. Example:

$$6.022 \times 10^{23} \text{ mol}^{-1}$$

4. When reporting results, match the power of ten in both numerator and denominator if possible, to minimize confusion.

Examples

- The radius of an electron: $2.82 \times 10^{-15} \text{ m}$
- Avogadro's number: $6.022 \times 10^{23} \text{ mol}^{-1}$
- Speed of light: $2.998 \times 10^8 \text{ m/s}$

3 Procedure to Plot a Good Graph

1. **Choose axes:** Put the independent variable on the x-axis and the dependent variable on the y-axis. Use SI units.
2. **Select a suitable scale:** Use simple, evenly spaced tick increments (1, 2, 5, 10, ...). Make full use of the graph area so the data spread occupies most of the plotting region.
3. **Label axes:** Always show the quantity and unit, e.g. V (V) or I (A).
4. **Plot points accurately:** Mark each data point clearly (dot or small cross). Avoid large symbols that obscure accuracy.
5. **Best-fit line or curve:** For linear relationships, draw a best-fit straight line (not a line joining the points). For nonlinear data, draw a smooth curve that follows the trend.
6. **Show uncertainties:** Where applicable, draw error bars representing measurement uncertainty.
7. **Extract quantities:** Determine slope, intercept, and uncertainties from the graph. Use $\text{slope} = \Delta y / \Delta x$ for linear fits.
8. **Neat presentation:** Use a ruler for axes and lines, include a title, and keep the graph free of unnecessary clutter.

Ohm's Law Example

Ohm's law states that $V = IR$, where V is the potential difference across a conductor, I is the current through it, and R is the resistance (assumed constant for an ohmic conductor). In a graph of current (I) versus voltage (V), the slope gives $\frac{\Delta I}{\Delta V} = 1/R$.

Sample measurements

The following set of measurements for voltage and current are recorded for a solar panel in Table. 1.

Voltage V (V)	Current I (mA)	Power $P = VI$ (mW)
0.63	4.0	0
1.05	3.6	3.6
...	...	...
2.85	0	0

Table 1: Measured voltage and current for solar panel.

Plot

The plot in Figure. 1 shows the data points and the best-fit line.

1. The best fit line should be a smooth line with points on either side. It need not pass through all the points.
2. We need to maximizeTo maximize the utilization of graph, a proper scale should be chosen. Consider the 1 cm squares as bricks of building material and you need to build a home on the available plot. Draw the axes two bricks inside to avoid punching problems. After drawing the axes, count the available land area in units of bricks along X and Y. Lets say we have 21 bricks along X and 16 along Y.
 - (a) Along X, find the range of data to be plotted. Lets say for voltage, we have data range from 0 V to 2.85 V. We can approximate range to $[0 \text{ V}, 3 \text{ V}]$. So the brick size is $3/18 = 0.16 \text{ V}$. For convenience, we can approximate it to 0.15 V.
 - (b) Along Y, find the range of data to be plotted. Lets say for current, we have data range from 0 V to 4 mA. So the brick size is $4/16 = 0.25 \text{ mA}$.

How to choose proper scale?

Aim is to maximize utilization of graph area.

Graph has 21 (1 cm) bricks along X and 16 along Y.

Quantity	Range	#Bricks	Brick-size
Voltage	[0 V, 2.85 V]	21	
	~[0 V, 3 V]	~18	$(3-0)/18 = 0.16 \text{ V}$ ~ 0.15 V
Current	[0 mA, 4 mA]	16	$(4-0)/16 = 0.25 \text{ mA}$
Power	[0 mW, 6.7 mW] ~[0 mW, 8 mW]	16	$(8-0)/16 = 0.5 \text{ mW}$

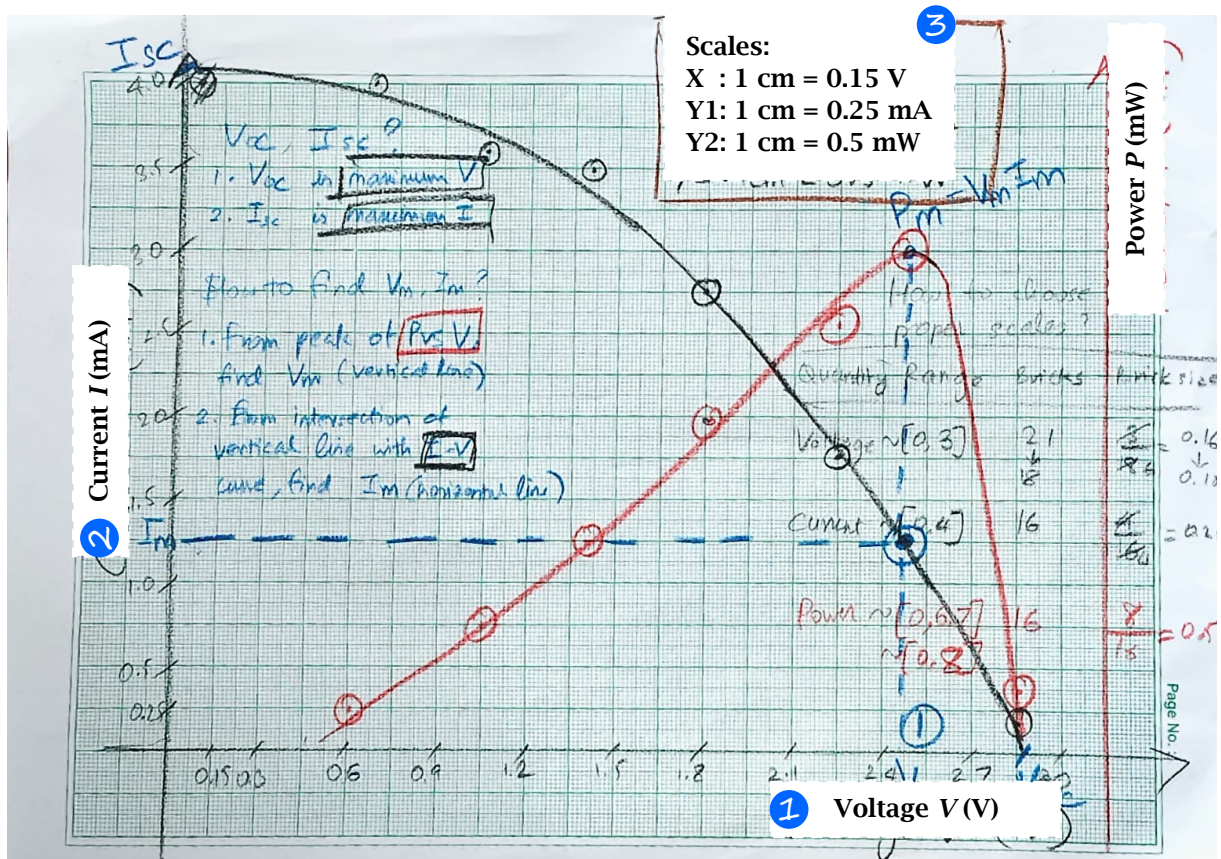


Figure 1: Solar cell practice.

Advanced Laboratory protocols

4 Finding Constants of a Straight Line Using Error Analysis

When two physical quantities are related linearly, their relation can be expressed as:

$$y = mx + c$$

where

- m is the slope (constant of proportionality),
- c is the intercept on the y -axis,
- x and y are the measured variables.

4.1 Experimental Data

During the experiment, pairs of values (x_i, y_i) are recorded. Each measurement has an associated error:

- Δx_i : least count or uncertainty in measurement of x ,
- Δy_i : least count or uncertainty in measurement of y .

4.2 Best-Fit Straight Line

A graph of y vs. x is plotted. The slope m and intercept c can be determined using the **method of least squares**:

$$m = \frac{N \sum x_i y_i - \sum x_i \sum y_i}{N \sum x_i^2 - (\sum x_i)^2}$$
$$c = \frac{\sum y_i \sum x_i^2 - \sum x_i \sum x_i y_i}{N \sum x_i^2 - (\sum x_i)^2}$$

where N is the number of data points.

Expressions in Terms of Averages

Let

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$$

be the mean values of the data. Then, the slope m and intercept c are given by:

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad c = \bar{y} - m\bar{x}.$$

4.3 Error in Slope and Intercept

The deviations of observed points from the line are:

$$d_i = y_i - (mx_i + c)$$

The sum of squared deviations is used to estimate the uncertainty in slope and intercept.
The standard error of estimate is:

$$\sigma = \sqrt{\frac{\sum d_i^2}{N - 2}}$$

The error in slope is given by:

$$\Delta m = \sigma \sqrt{\frac{N}{N \sum x_i^2 - (\sum x_i)^2}}$$

The error in intercept is:

$$\Delta c = \sigma \sqrt{\frac{\sum x_i^2}{N \sum x_i^2 - (\sum x_i)^2}}$$

5 Significant Figures

In experimental physics, every measured quantity has an associated **uncertainty**. The number of **significant figures** used in reporting results must reflect this uncertainty.

1. **Definition:** Significant figures are the digits in a number that carry meaningful information about its precision. This includes all certain digits and the first uncertain digit.
2. **Rules for Use:**
 - All non-zero digits are significant.
 - Zeros between non-zero digits are significant.
 - Leading zeros are not significant; they only indicate the position of the decimal point.
 - Trailing zeros after a decimal point are significant.
3. **In Calculations:**
 - For multiplication or division: the result should have as many significant figures as the measured value with the *least number* of significant figures.
 - For addition or subtraction: the result should be rounded off to the *least precise decimal place* among the quantities involved.
4. **In Reporting Final Results:**
 - The uncertainty (error) should be quoted to *one significant figure* (occasionally two if the first digit is 1 or 2).
 - The measured value should be reported up to the same decimal place as the uncertainty.

Example: Counting Significant Figures

- 1234 $\rightarrow$ 4 significant figures (all non-zero digits).
- 0.00456 $\rightarrow$ 3 significant figures (leading zeros are not significant).
- 2.300 $\rightarrow$ 4 significant figures (trailing zeros after the decimal point are significant).
- 7.0890 $\rightarrow$ 5 significant figures (zeros between non-zero digits and the final zero are significant).
- 1.20×10^3 $\rightarrow$ 3 significant figures (scientific notation helps to clearly indicate them).

Example:

If resistance is found to be

$$R = 12.36 \, \Omega \pm 0.07 \, \Omega,$$

the correct way to report is

$$R = 12.36 \pm 0.07 \, \Omega$$

and not 12.360 ± 0.07 , since extra digits suggest unwarranted precision.

5.1 Final Result

The constants of the line are written with their uncertainties as:

$$m \pm \Delta m, \quad c \pm \Delta c$$

This ensures the linear relation is reported within the limits of experimental accuracy.

Note: This method is commonly applied in laboratory experiments such as verification of Ohm's law, Hooke's law, and determination of refractive index, where linear relations are analyzed.

Notes and Tips

- Always record data in scientific notation when the number is too large or too small for convenient decimal representation.
- Always include units and report final values with **appropriate significant figures** and uncertainty.